CORE PURE / DIFFERENTIAL EQUATIONS

2nd Order Differential Equations

Worked solutions 12 questions · 125 marks

Questions

Question 1	*6 marks*	2
Question 2	*11 marks*	4
Question 3	*11 marks*	7
Question 4	*11 marks*	10
Question 5	*7 marks*	12
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Question 12	*10 marks*	32

1. **(a)** Find the general solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4y = 0 \qquad \textbf{[2]}$$

(b) Hence, by considering a particular integral of the form $x(A\cos 2x + B\sin 2x)$, find the general solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4y = 8\cos 2x - 12\sin 2x \qquad \textbf{[4]}$$

Solution

(a) For

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4y = 0$$

assume a solution of the form $y = e^{\lambda x}$. Then the auxiliary equation is

$$\lambda^2 + 4 = 0$$

which has roots $\lambda = \pm 2i$. Hence the general solution is

$$y = C_1\cos 2x + C_2\sin 2x$$

(b) From part (a), the complementary function is

$$y_c = C_1\cos 2x + C_2\sin 2x$$

Since the right-hand side involves $\cos 2x$ and $\sin 2x$, take a particular integral in the given form

$$y_p = x(A\cos 2x + B\sin 2x)$$

Differentiate:

$$\frac{\mathrm{d}y_p}{\mathrm{d}x} = (A\cos 2x + B\sin 2x) + x(-2A\sin 2x + 2B\cos 2x)$$

Differentiate again:

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} &= (-2A\sin 2x + 2B\cos 2x) + (-2A\sin 2x + 2B\cos 2x)\\ &\quad + x(-4A\cos 2x - 4B\sin 2x)\\ &= -4A\sin 2x + 4B\cos 2x - 4x(A\cos 2x + B\sin 2x)\end{aligned}$$

Now substitute into $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4y$:

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} + 4y_p &= [-4A\sin 2x + 4B\cos 2x - 4x(A\cos 2x + B\sin 2x)]\\ &\quad + 4x(A\cos 2x + B\sin 2x)\\ &= 4B\cos 2x - 4A\sin 2x\end{aligned}$$

This must equal the right-hand side:

$$4B\cos 2x - 4A\sin 2x = 8\cos 2x - 12\sin 2x$$

Equating coefficients gives

$$4B = 8$$

$$-4A = -12$$

so

$$B = 2, \qquad A = 3$$

Therefore

$$y_p = x(3\cos 2x + 2\sin 2x)$$

Hence the general solution is

$$y = y_c + y_p = C_1 \cos 2x + C_2 \sin 2x + x(3\cos 2x + 2\sin 2x)$$

2. Find the particular solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 5y = 8\cos t$$

given that, when $t = 0$, $y = 3$ and $\frac{\mathrm{d}y}{\mathrm{d}t} = 1$. **[11]**

Solution

First solve the associated homogeneous equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 5y = 0$$

Its auxiliary equation is

$$\lambda^2 + 4\lambda + 5 = 0$$

which has roots $\lambda = -2 \pm i$. Hence the complementary function is

$$y_c = e^{-2t}(A\cos t + B\sin t)$$

Now find a particular integral for

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 5y = 8\cos t$$

Since the right-hand side is $8\cos t$, try

$$y_p = a\cos t + b\sin t$$

Then

$$\frac{\mathrm{d}y_p}{\mathrm{d}t} = -a\sin t + b\cos t$$

and

$$\frac{\mathrm{d}^2y_p}{\mathrm{d}t^2} = -a\cos t - b\sin t$$

Substituting into the differential equation:

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y_p}{\mathrm{d}t} + 5y_p &= (-a\cos t - b\sin t) + 4(-a\sin t + b\cos t) + 5(a\cos t + b\sin t)\\ &= (-a + 4b + 5a)\cos t + (-b - 4a + 5b)\sin t\\ &= (4a + 4b)\cos t + (-4a + 4b)\sin t\end{aligned}$$

This must equal $8\cos t$, so equating coefficients gives

$$4a + 4b = 8, \qquad -4a + 4b = 0$$

Divide both equations by 4:

$$a + b = 2, \qquad -a + b = 0$$

From $-a + b = 0$, we get

$$b = a$$

Substitute into $a + b = 2$:

$$a + a = 2$$

$$2a = 2$$

$$a = 1$$

Therefore

$$b = 1$$

So the particular integral is

$$y_p = \cos t + \sin t$$

Hence the general solution is

$$y = e^{-2t}(A\cos t + B\sin t) + \cos t + \sin t$$

Now use the initial conditions.

When $t = 0$, $y = 3$:

$$\begin{aligned} 3 &= e^0(A\cos 0 + B\sin 0) + \cos 0 + \sin 0 \\ &= 1(A \times 1 + B \times 0) + 1 + 0 \\ &= A + 1 \end{aligned}$$

So

$$A + 1 = 3$$

$$A = 2$$

Next differentiate the general solution:

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}t} &= \frac{\mathrm{d}}{\mathrm{d}t}\left[e^{-2t}(A\cos t + B\sin t)\right] + \frac{\mathrm{d}}{\mathrm{d}t}(\cos t + \sin t) \\ &= -2e^{-2t}(A\cos t + B\sin t) + e^{-2t}(-A\sin t + B\cos t) - \sin t + \cos t \\ &= e^{-2t}[(-2A + B)\cos t + (-A - 2B)\sin t] - \sin t + \cos t \end{aligned}$$

When $t = 0$, $\frac{\mathrm{d}y}{\mathrm{d}t} = 1$:

$$\begin{aligned} 1 &= e^0[(-2A + B)\cos 0 + (-A - 2B)\sin 0] - \sin 0 + \cos 0 \\ &= 1[(-2A + B) \times 1 + (-A - 2B) \times 0] - 0 + 1 \\ &= (-2A + B) + 1 \end{aligned}$$

So

$$(-2A + B) + 1 = 1$$

$$-2A + B = 0$$

Using $A = 2$:

$$-2(2) + B = 0$$

$$-4 + B = 0$$

$$B = 4$$

Therefore the particular solution satisfying the given conditions is

$$y = e^{-2t}(2\cos t + 4\sin t) + \cos t + \sin t$$

3. A damped oscillator is governed by

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 6\frac{\mathrm{d}y}{\mathrm{d}t} + 13y = 75\sin 2t$$

(a) Find the general solution of this differential equation. **[7]**

(b) Show that, whatever the initial conditions, the motion is approximately of the form

$$y \approx R\sin(2t - \phi)$$

for large positive values of t, where $R > 0$ and $0 < \phi < \frac{\pi}{2}$. Determine R and ϕ. **[4]**

Solution

(a) First solve the associated homogeneous equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 6\frac{\mathrm{d}y}{\mathrm{d}t} + 13y = 0$$

Using the trial solution $y = e^{\lambda t}$ gives the auxiliary equation

$$\lambda^2 + 6\lambda + 13 = 0$$

which has roots $\lambda = -3 \pm 2i$. Hence the complementary function is

$$y_c = e^{-3t}(A\cos 2t + B\sin 2t)$$

For a particular integral, try

$$y_p = a\sin 2t + b\cos 2t$$

Then

$$\frac{\mathrm{d}y_p}{\mathrm{d}t} = 2a\cos 2t - 2b\sin 2t$$

$$\frac{\mathrm{d}^2y_p}{\mathrm{d}t^2} = -4a\sin 2t - 4b\cos 2t$$

Substituting into the differential equation:

$$(-4a\sin 2t - 4b\cos 2t) + 6(2a\cos 2t - 2b\sin 2t) + 13(a\sin 2t + b\cos 2t)$$
$$= 75\sin 2t$$

Collecting $\sin 2t$ and $\cos 2t$ terms:

$$(-4a - 12b + 13a)\sin 2t + (12a - 4b + 13b)\cos 2t$$
$$= 75\sin 2t$$

So

$$(9a - 12b)\sin 2t + (12a + 9b)\cos 2t = 75\sin 2t$$

Equating coefficients gives

$$9a - 12b = 75$$
$$12a + 9b = 0$$

Solve these simultaneously. From the second equation:

$$4a + 3b = 0 \quad \Longrightarrow \quad b = -\frac{4a}{3}$$

Substitute into the first:

$$9a - 12\left(-\frac{4a}{3}\right) = 75$$
$$9a + 16a = 75$$
$$25a = 75$$
$$a = 3$$

Then

$$b = -\frac{4(3)}{3} = -4$$

Thus

$$y_p = 3\sin 2t - 4\cos 2t$$

Therefore the general solution is

$$y = e^{-3t}(A\cos 2t + B\sin 2t) + 3\sin 2t - 4\cos 2t$$

(b) From part (a):

$$y = e^{-3t}(A\cos 2t + B\sin 2t) + 3\sin 2t - 4\cos 2t$$

As $t \to +\infty$, the exponential factor tends to 0. Also $\sin 2t$ and $\cos 2t$ remain bounded, so

$$e^{-3t}(A\cos 2t + B\sin 2t) \to 0$$

Therefore, whatever the initial conditions, for large positive t:

$$y \approx 3\sin 2t - 4\cos 2t$$

Now write this in the form $R\sin(2t - \phi)$. Using the identity

$$\sin(2t - \phi) = \sin 2t\cos\phi - \cos 2t\sin\phi$$

we have

$$R\sin(2t - \phi) = R\cos\phi\ \sin 2t - R\sin\phi\ \cos 2t$$

Equating coefficients with $3\sin 2t - 4\cos 2t$ gives

$$R\cos\phi = 3$$
$$R\sin\phi = 4$$

Squaring and adding:

$$R^2\cos^2\phi + R^2\sin^2\phi = 3^2 + 4^2$$
$$R^2(\cos^2\phi + \sin^2\phi) = 25$$
$$R^2 = 25$$

Since $R > 0$:

$$R = 5$$

Also

$$\tan\phi = \frac{R\sin\phi}{R\cos\phi} = \frac{4}{3}$$

Since $0 < \phi < \frac{\pi}{2}$:

$$\phi = \arctan\left(\frac{4}{3}\right) \approx 0.927$$

Hence, for large positive t:

$$y \approx 5\sin\left(2t - \arctan\frac{4}{3}\right)$$

So the required constants are $R = 5$ and $\phi = \arctan\left(\frac{4}{3}\right) \approx 0.927$.

4. A function $y(t)$ satisfies the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\frac{\mathrm{d}y}{\mathrm{d}t} + 5y = 10\cos t$$

Given that, when $t = 0$, $y = 5$ and $\frac{\mathrm{d}y}{\mathrm{d}t} = 2$, find the particular solution. **[11]**

Solution

First solve the associated homogeneous equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\frac{\mathrm{d}y}{\mathrm{d}t} + 5y = 0$$

Its auxiliary equation is

$$\lambda^2 + 2\lambda + 5 = 0$$

which has roots $\lambda = -1 \pm 2i$. So the complementary function is

$$y_c = e^{-t}(A\cos 2t + B\sin 2t)$$

Now find a particular integral for

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\frac{\mathrm{d}y}{\mathrm{d}t} + 5y = 10\cos t$$

Since the right-hand side is $10\cos t$, try

$$y_p = a\cos t + b\sin t$$

Then

$$\frac{\mathrm{d}y_p}{\mathrm{d}t} = -a\sin t + b\cos t$$

and

$$\frac{\mathrm{d}^2y_p}{\mathrm{d}t^2} = -a\cos t - b\sin t$$

Substituting into the differential equation:

$$(-a\cos t - b\sin t) + 2(-a\sin t + b\cos t) + 5(a\cos t + b\sin t) = 10\cos t$$

Collecting $\cos t$ and $\sin t$ terms:

$$(-a + 2b + 5a)\cos t + (-b - 2a + 5b)\sin t = 10\cos t$$

$$(4a + 2b)\cos t + (4b - 2a)\sin t = 10\cos t$$

Equating coefficients gives

$$4a + 2b = 10, \qquad 4b - 2a = 0$$

From $4b - 2a = 0$:

$$a = 2b$$

Substitute into $4a + 2b = 10$:

$$\begin{aligned} 4(2b) + 2b &= 10 \\ 8b + 2b &= 10 \\ 10b &= 10 \\ b &= 1 \end{aligned}$$

Hence

$$a = 2$$

So the particular integral is

$$y_p = 2\cos t + \sin t$$

Therefore the general solution is

$$y = e^{-t}(A\cos 2t + B\sin 2t) + 2\cos t + \sin t$$

Now use the initial conditions.

When $t = 0$, $y = 5$:

$$\begin{aligned} 5 &= e^0(A\cos 0 + B\sin 0) + 2\cos 0 + \sin 0 \\ &= A + 2 \end{aligned}$$

So

$$A = 3$$

Next differentiate the general solution:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left[e^{-t}(A\cos 2t + B\sin 2t)\right] + \frac{\mathrm{d}}{\mathrm{d}t}(2\cos t + \sin t)$$

Using the product rule on the first term:

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}t} &= -e^{-t}(A\cos 2t + B\sin 2t) + e^{-t}(-2A\sin 2t + 2B\cos 2t) - 2\sin t + \cos t \\ &= e^{-t}[(-A + 2B)\cos 2t + (-2A - B)\sin 2t] - 2\sin t + \cos t \end{aligned}$$

When $t = 0$, $\frac{\mathrm{d}y}{\mathrm{d}t} = 2$:

$$\begin{aligned} 2 &= e^0[(-A + 2B)\cos 0 + (-2A - B)\sin 0] - 2\sin 0 + \cos 0 \\ &= (-A + 2B) + 1 \end{aligned}$$

Substitute $A = 3$:

$$\begin{aligned} 2 &= -3 + 2B + 1 \\ 2 &= 2B - 2 \\ 4 &= 2B \\ B &= 2 \end{aligned}$$

Hence the required particular solution satisfying the given conditions is

$$y(t) = e^{-t}(3\cos 2t + 2\sin 2t) + 2\cos t + \sin t$$

5. **(a)** Find the general solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 13y = 0 \qquad \textbf{[2]}$$

(b) Hence find the general solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 13y = e^{-2x}(6\cos x - 8\sin x) \qquad \textbf{[5]}$$

Solution

(a) For the homogeneous equation, use the auxiliary equation

$$\lambda^2 + 4\lambda + 13 = 0$$

which has roots $\lambda = -2 \pm 3i$. Hence the general solution is

$$y = e^{-2x}(A\cos 3x + B\sin 3x)$$

(b) From part (a), the complementary function is

$$y_c = e^{-2x}(A\cos 3x + B\sin 3x)$$

For a particular integral, try

$$y_p = e^{-2x}(a\cos x + b\sin x)$$

Then

$$\begin{aligned}\frac{\mathrm{d}y_p}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[e^{-2x}(a\cos x + b\sin x)\right] \\ &= -2e^{-2x}(a\cos x + b\sin x) + e^{-2x}(-a\sin x + b\cos x) \\ &= e^{-2x}[(b-2a)\cos x + (-a-2b)\sin x]\end{aligned}$$

Also

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} &= \frac{\mathrm{d}}{\mathrm{d}x}\left\{e^{-2x}[(b-2a)\cos x + (-a-2b)\sin x]\right\} \\ &= -2e^{-2x}[(b-2a)\cos x + (-a-2b)\sin x] \\ &\quad + e^{-2x}[(-b+2a)\sin x + (-a-2b)\cos x] \\ &= e^{-2x}[(3a-4b)\cos x + (4a+3b)\sin x]\end{aligned}$$

Substituting into

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 13y = e^{-2x}(6\cos x - 8\sin x)$$

gives

$$\begin{aligned} & e^{-2x}[(3a-4b)\cos x+(4a+3b)\sin x] \\ & \quad +4e^{-2x}[(b-2a)\cos x+(-a-2b)\sin x] \\ & \quad +13e^{-2x}(a\cos x+b\sin x) \\ & =e^{-2x}(6\cos x-8\sin x) \end{aligned}$$

So

$$e^{-2x}(8a\cos x+8b\sin x)=e^{-2x}(6\cos x-8\sin x)$$

Hence

$$8a\cos x+8b\sin x=6\cos x-8\sin x$$

Equating coefficients gives

$$8a=6, \qquad 8b=-8$$

So

$$a=\frac{3}{4}, \qquad b=-1$$

Therefore

$$y_p=e^{-2x}\left(\frac{3}{4}\cos x-\sin x\right)$$

Thus the general solution is

$$y=e^{-2x}(A\cos 3x+B\sin 3x)+e^{-2x}\left(\frac{3}{4}\cos x-\sin x\right)$$

So

$$y=e^{-2x}\left(A\cos 3x+B\sin 3x+\frac{3}{4}\cos x-\sin x\right)$$

6. Find the particular solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 13y = 28\cos 3t - 4\sin 3t$$

given that, when $t = 0$, $y = 5$ and $\frac{\mathrm{d}y}{\mathrm{d}t} = 7$. **[11]**

Solution

We solve

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 13y = 28\cos 3t - 4\sin 3t$$

by finding the complementary function and a particular integral, then using the initial conditions.

For the complementary function, solve the auxiliary equation

$$\lambda^2 + 4\lambda + 13 = 0$$

which has roots $\lambda = -2 \pm 3i$. Hence the complementary function is

$$y_c = e^{-2t}(A\cos 3t + B\sin 3t)$$

For the particular integral, since the right-hand side is of the form $28\cos 3t - 4\sin 3t$, try

$$y_p = a\cos 3t + b\sin 3t$$

Then

$$\frac{\mathrm{d}y_p}{\mathrm{d}t} = -3a\sin 3t + 3b\cos 3t$$

$$\frac{\mathrm{d}^2y_p}{\mathrm{d}t^2} = -9a\cos 3t - 9b\sin 3t$$

Substituting into the differential equation:

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y_p}{\mathrm{d}t} + 13y_p &= (-9a\cos 3t - 9b\sin 3t) + 4(-3a\sin 3t + 3b\cos 3t) + 13(a\cos 3t + b\sin 3t) \\ &= (-9a + 12b + 13a)\cos 3t + (-9b - 12a + 13b)\sin 3t \\ &= (4a + 12b)\cos 3t + (4b - 12a)\sin 3t\end{aligned}$$

Equating coefficients with $28\cos 3t - 4\sin 3t$ gives

$$4a + 12b = 28$$

$$4b - 12a = -4$$

Dividing both equations by 4:

$$a + 3b = 7$$

$$b - 3a = -1$$

From the second equation:

$$b = 3a - 1$$

Substitute into the first:

$$\begin{aligned} a + 3(3a - 1) &= 7 \\ a + 9a - 3 &= 7 \\ 10a &= 10 \\ a &= 1 \end{aligned}$$

Then

$$b = 3(1) - 1 = 2$$

So

$$y_p = \cos 3t + 2 \sin 3t$$

Hence the general solution is

$$y = e^{-2t}(A \cos 3t + B \sin 3t) + \cos 3t + 2 \sin 3t$$

Now use the initial conditions.

First, when $t = 0$, $y = 5$:

$$\begin{aligned} 5 &= e^0(A \cos 0 + B \sin 0) + \cos 0 + 2 \sin 0 \\ &= A(1) + B(0) + 1 + 0 \\ &= A + 1 \end{aligned}$$

So

$$A = 4$$

Next differentiate y, starting with the complementary function:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\left[e^{-2t}(A \cos 3t + B \sin 3t)\right] &= e^{-2t}(-2)(A \cos 3t + B \sin 3t) + e^{-2t}(-3A \sin 3t + 3B \cos 3t) \\ &= e^{-2t}[(3B - 2A) \cos 3t + (-3A - 2B) \sin 3t] \end{aligned}$$

Also

$$\frac{\mathrm{d}}{\mathrm{d}t}(\cos 3t + 2 \sin 3t) = -3 \sin 3t + 6 \cos 3t$$

Therefore

$$\frac{\mathrm{d}y}{\mathrm{d}t} = e^{-2t}[(3B - 2A) \cos 3t + (-3A - 2B) \sin 3t] - 3 \sin 3t + 6 \cos 3t$$

When $t = 0$, $\frac{\mathrm{d}y}{\mathrm{d}t} = 7$:

$$\begin{aligned} 7 &= e^0[(3B - 2A) \cos 0 + (-3A - 2B) \sin 0] - 3 \sin 0 + 6 \cos 0 \\ &= (3B - 2A)(1) + 0 + 0 + 6 \\ &= 3B - 2A + 6 \end{aligned}$$

Substitute $A = 4$:

$$\begin{aligned} 7 &= 3B - 8 + 6 \\ 7 &= 3B - 2 \\ 3B &= 9 \\ B &= 3 \end{aligned}$$

Hence the required particular solution is

$$y = e^{-2t}(4\cos 3t + 3\sin 3t) + \cos 3t + 2\sin 3t$$

7. Solve the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} - 6y = 50\sin x$$

subject to the conditions $y(0) = 3$ and $y'(0) = -6$. **[11]**

Solution

We solve

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} - 6y = 50\sin x$$

subject to

$$y(0) = 3, \qquad y'(0) = -6$$

First, solve the homogeneous equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} - 6y = 0$$

Its auxiliary equation is

$$\lambda^2 - \lambda - 6 = 0$$

Factorising:

$$(\lambda - 3)(\lambda + 2) = 0$$

So the roots are $\lambda = 3$ and $\lambda = -2$. Hence the complementary function is

$$y_c = Ae^{3x} + Be^{-2x}$$

Now find a particular integral. Since the right-hand side is $50\sin x$, try

$$y_p = a\sin x + b\cos x$$

Then

$$\frac{\mathrm{d}y_p}{\mathrm{d}x} = a\cos x - b\sin x$$

$$\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} = -a\sin x - b\cos x$$

Substituting into the differential equation:

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} - \frac{\mathrm{d}y_p}{\mathrm{d}x} - 6y_p &= (-a\sin x - b\cos x) - (a\cos x - b\sin x) - 6(a\sin x + b\cos x)\\ &= -a\sin x - b\cos x - a\cos x + b\sin x - 6a\sin x - 6b\cos x\\ &= (b - 7a)\sin x + (-a - 7b)\cos x\end{aligned}$$

We want this to equal $50\sin x$, so equating coefficients gives

$$b - 7a = 50$$

$$-a - 7b = 0$$

From the second equation:

$$a = -7b$$

Substitute into the first:

$$\begin{aligned} b - 7(-7b) &= 50 \\ b + 49b &= 50 \\ 50b &= 50 \\ b &= 1 \end{aligned}$$

So

$$a = -7$$

Therefore the particular integral is

$$y_p = -7\sin x + \cos x$$

Hence the general solution is

$$y = Ae^{3x} + Be^{-2x} - 7\sin x + \cos x$$

Now use the initial conditions. First, substitute $x = 0$:

$$\begin{aligned} y(0) &= Ae^0 + Be^0 - 7\sin 0 + \cos 0 \\ &= A + B + 1 \end{aligned}$$

Since $y(0) = 3$:

$$A + B + 1 = 3$$

so

$$A + B = 2$$

Next differentiate the general solution:

$$y'(x) = 3Ae^{3x} - 2Be^{-2x} - 7\cos x - \sin x$$

Now apply $y'(0) = -6$:

$$\begin{aligned} y'(0) &= 3Ae^0 - 2Be^0 - 7\cos 0 - \sin 0 \\ &= 3A - 2B - 7 \end{aligned}$$

So

$$3A - 2B - 7 = -6$$

hence

$$3A - 2B = 1$$

We now solve

$$\begin{aligned} A + B &= 2 \\ 3A - 2B &= 1 \end{aligned}$$

Multiply the first equation by 2:

$$2A + 2B = 4$$

Add this to $3A - 2B = 1$:

$$\begin{aligned}(2A + 2B) + (3A - 2B) &= 4 + 1\\ 5A &= 5\\ A &= 1\end{aligned}$$

Then from $A + B = 2$:

$$\begin{aligned}1 + B &= 2\\ B &= 1\end{aligned}$$

Therefore the required solution is

$$y = e^{3x} + e^{-2x} - 7\sin x + \cos x$$

8. Find the particular solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 13y = 8e^{-2t}\sin t$$

given that, when $t = 0$, $y = 1$ and $\frac{\mathrm{d}y}{\mathrm{d}t} = 2$. **[11]**

Solution

The homogeneous equation is

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 13y = 0$$

The auxiliary equation is

$$\lambda^2 + 4\lambda + 13 = 0$$

which has roots $\lambda = -2 \pm 3i$. Hence

$$y_c = e^{-2t}(A\cos 3t + B\sin 3t)$$

For a particular integral, try

$$y_p = e^{-2t}(a\sin t + b\cos t)$$

Then

$$\begin{aligned}\frac{\mathrm{d}y_p}{\mathrm{d}t} &= \frac{\mathrm{d}}{\mathrm{d}t}\left[e^{-2t}(a\sin t + b\cos t)\right] \\ &= -2e^{-2t}(a\sin t + b\cos t) + e^{-2t}(a\cos t - b\sin t) \\ &= e^{-2t}[(-b-2a)\sin t + (a-2b)\cos t]\end{aligned}$$

Also

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}t^2} &= \frac{\mathrm{d}}{\mathrm{d}t}\left\{e^{-2t}[(-b-2a)\sin t + (a-2b)\cos t]\right\} \\ &= -2e^{-2t}[(-b-2a)\sin t + (a-2b)\cos t] \\ &\quad + e^{-2t}[(-b-2a)\cos t - (a-2b)\sin t] \\ &= e^{-2t}[(3a+4b)\sin t + (-4a+3b)\cos t]\end{aligned}$$

Substituting into

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 13y = 8e^{-2t}\sin t$$

gives

$$\begin{aligned}&e^{-2t}[(3a+4b)\sin t + (-4a+3b)\cos t] \\ &\quad + 4e^{-2t}[(-b-2a)\sin t + (a-2b)\cos t] \\ &\quad + 13e^{-2t}(a\sin t + b\cos t) \\ &= 8e^{-2t}\sin t\end{aligned}$$

So

$$e^{-2t}(8a\sin t + 8b\cos t) = 8e^{-2t}\sin t$$

Hence

$$8a\sin t + 8b\cos t = 8\sin t$$

Equating coefficients gives

$$8a = 8, \qquad 8b = 0$$

So

$$a = 1, \qquad b = 0$$

Therefore

$$y_p = e^{-2t}\sin t$$

Thus the general solution is

$$y = e^{-2t}(A\cos 3t + B\sin 3t) + e^{-2t}\sin t$$

So

$$y = e^{-2t}(A\cos 3t + B\sin 3t + \sin t)$$

Now use the initial conditions.

When $t = 0$, $y = 1$:

$$1 = e^{0}(A\cos 0 + B\sin 0 + \sin 0)$$

So

$$1 = A$$

Hence

$$A = 1$$

Now differentiate:

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}t} &= \frac{\mathrm{d}}{\mathrm{d}t}\left[e^{-2t}(A\cos 3t + B\sin 3t + \sin t)\right] \\ &= -2e^{-2t}(A\cos 3t + B\sin 3t + \sin t) \\ &\quad + e^{-2t}(-3A\sin 3t + 3B\cos 3t + \cos t) \\ &= e^{-2t}(-3A\sin 3t + 3B\cos 3t + \cos t - 2A\cos 3t - 2B\sin 3t - 2\sin t)\end{aligned}$$

When $t = 0$, $\frac{\mathrm{d}y}{\mathrm{d}t} = 2$, so

$$2 = 3B + 1 - 2A$$

With $A = 1$:

$$2 = 3B + 1 - 2$$

So

$$2 = 3B - 1$$

Hence

$$3B = 3$$

So

$$B = 1$$

Substituting $A = 1$ and $B = 1$, the required particular solution is

$$y = e^{-2t}\big(\cos 3t + \sin 3t + \sin t\big)$$

9. **(a)** Show that, for all real x,

$$(1+\tanh x)\cosh x = e^x$$ **[2]**

(b) Hence solve the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 2\frac{\mathrm{d}y}{\mathrm{d}x} + y = 3(1+\tanh x)\cosh x$$

given that, when $x = 0$, $y = 1$ and $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$. **[10]**

Solution

(a) Using

$$\tanh x = \frac{\sinh x}{\cosh x}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}, \qquad \sinh x = \frac{e^x - e^{-x}}{2}$$

we have

$$\begin{aligned}(1+\tanh x)\cosh x &= \left(1 + \frac{\sinh x}{\cosh x}\right)\cosh x \\ &= \cosh x + \sinh x \\ &= \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2} \\ &= e^x\end{aligned}$$

Hence, for all real x:

$$(1+\tanh x)\cosh x = e^x$$

(b) From part (a), the differential equation becomes

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 2\frac{\mathrm{d}y}{\mathrm{d}x} + y = 3e^x$$

The homogeneous equation is

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 2\frac{\mathrm{d}y}{\mathrm{d}x} + y = 0$$

The auxiliary equation is

$$\lambda^2 - 2\lambda + 1 = 0$$

So

$$(\lambda - 1)^2 = 0$$

Hence the complementary function is

$$y_c = (A + Bx)e^x$$

For a particular integral, since the right-hand side is $3e^x$ and $\lambda = 1$ is a repeated root, try

$$y_p = ax^2e^x$$

Then

$$\begin{aligned}\frac{\mathrm{d}y_p}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left(ax^2e^x\right)\\ &= 2axe^x + ax^2e^x\\ &= e^x\left(ax^2 + 2ax\right)\end{aligned}$$

Differentiate again:

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[e^x\left(ax^2 + 2ax\right)\right]\\ &= e^x\left(ax^2 + 2ax\right) + e^x(2ax + 2a)\\ &= e^x\left(ax^2 + 4ax + 2a\right)\end{aligned}$$

Substituting into the differential equation gives

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} - 2\frac{\mathrm{d}y_p}{\mathrm{d}x} + y_p &= e^x\left(ax^2 + 4ax + 2a\right) - 2e^x\left(ax^2 + 2ax\right) + ax^2e^x\\ &= 2ae^x\end{aligned}$$

So

$$2ae^x = 3e^x$$

Hence

$$2a = 3$$

So

$$a = \frac{3}{2}$$

Therefore

$$y_p = \frac{3x^2}{2}e^x$$

Thus the general solution is

$$y = (A + Bx)e^x + \frac{3x^2}{2}e^x$$

So

$$y = e^x\left(A + Bx + \frac{3x^2}{2}\right)$$

Now use the initial conditions.

When $x = 0$, $y = 1$:

$$\begin{aligned}1 &= e^0\left(A + B \times 0 + \frac{3(0)^2}{2}\right)\\ &= A\end{aligned}$$

So

$$A = 1$$

Now differentiate:

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[e^x\left(A + Bx + \frac{3x^2}{2}\right)\right] \\ &= e^x\left(A + Bx + \frac{3x^2}{2}\right) + e^x(B + 3x) \\ &= e^x\left(A + Bx + \frac{3x^2}{2} + B + 3x\right)\end{aligned}$$

When $x = 0$, $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$, so

$$0 = A + B$$

With $A = 1$:

$$0 = 1 + B$$

Hence

$$B = -1$$

Substituting $A = 1$ and $B = -1$, the required solution is

$$y = e^x\left(\frac{3x^2}{2} - x + 1\right)$$

10. **(a)** Show that

$$\frac{\cosh x - \sinh x}{\cosh x + \sinh x} = e^{-2x}$$

[2]

(b) Hence find the particular solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 4\left(\frac{\cosh x - \sinh x}{\cosh x + \sinh x}\right)$$

given that, when $x = 0$, $y = 1$ and $\frac{\mathrm{d}y}{\mathrm{d}x} = 1$. **[10]**

Solution

(a) Using

$$\cosh x = \frac{e^x + e^{-x}}{2} \qquad \text{and} \qquad \sinh x = \frac{e^x - e^{-x}}{2}$$

we get

$$\begin{aligned}\cosh x - \sinh x &= \frac{e^x + e^{-x}}{2} - \frac{e^x - e^{-x}}{2} \\ &= \frac{e^x + e^{-x} - e^x + e^{-x}}{2} \\ &= \frac{2e^{-x}}{2} \\ &= e^{-x}\end{aligned}$$

Also

$$\begin{aligned}\cosh x + \sinh x &= \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2} \\ &= \frac{e^x + e^{-x} + e^x - e^{-x}}{2} \\ &= \frac{2e^x}{2} \\ &= e^x\end{aligned}$$

Therefore

$$\frac{\cosh x - \sinh x}{\cosh x + \sinh x} = \frac{e^{-x}}{e^x} = e^{-2x}$$

(b) From part (a), the differential equation becomes

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 4e^{-2x}$$

The homogeneous equation is

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 0$$

The auxiliary equation is

$$\lambda^2 + 4\lambda + 4 = 0$$

So

$$(\lambda + 2)^2 = 0$$

Hence the complementary function is

$$y_c = (A + Bx)e^{-2x}$$

Since the right-hand side is $4e^{-2x}$ and -2 is a repeated root of the auxiliary equation, try

$$y_p = ax^2e^{-2x}$$

Differentiate:

$$\begin{aligned}\frac{\mathrm{d}y_p}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left(ax^2e^{-2x}\right)\\ &= 2axe^{-2x} - 2ax^2e^{-2x}\\ &= e^{-2x}(2ax - 2ax^2)\end{aligned}$$

Differentiate again:

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[e^{-2x}(2ax - 2ax^2)\right]\\ &= -2e^{-2x}(2ax - 2ax^2) + e^{-2x}(2a - 4ax)\\ &= e^{-2x}(2a - 8ax + 4ax^2)\end{aligned}$$

Substituting into the differential equation gives

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y_p}{\mathrm{d}x} + 4y_p &= e^{-2x}(2a - 8ax + 4ax^2) + 4e^{-2x}(2ax - 2ax^2) + 4ax^2e^{-2x}\\ &= e^{-2x}(2a)\end{aligned}$$

So

$$2ae^{-2x} = 4e^{-2x}$$

Hence

$$2a = 4$$

So

$$a = 2$$

Therefore

$$y_p = 2x^2e^{-2x}$$

Thus the general solution is

$$y = (A + Bx)e^{-2x} + 2x^2e^{-2x}$$

So

$$y = e^{-2x}(2x^2 + Bx + A)$$

Now use the initial conditions.

When $x = 0$, $y = 1$:

$$\begin{aligned} 1 &= e^0(2 \times 0^2 + B \times 0 + A) \\ &= A \end{aligned}$$

So

$$A = 1$$

Now differentiate:

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[e^{-2x}(2x^2 + Bx + A)\right] \\ &= -2e^{-2x}(2x^2 + Bx + A) + e^{-2x}(4x + B) \\ &= e^{-2x}(4x + B - 4x^2 - 2Bx - 2A) \end{aligned}$$

When $x = 0$, $\frac{\mathrm{d}y}{\mathrm{d}x} = 1$, so

$$1 = B - 2A$$

With $A = 1$:

$$1 = B - 2$$

Hence

$$B = 3$$

Substituting $A = 1$ and $B = 3$, the required particular solution is

$$y = e^{-2x}(2x^2 + 3x + 1)$$

11. **(a)** Show that

$$3\cosh x - \sinh x = e^x + 2e^{-x}$$

[2]

(b) Hence solve the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} - 2y = 6(3\cosh x - \sinh x)$$

given that, when $x = 0$, $y = 1$ and $\frac{\mathrm{d}y}{\mathrm{d}x} = 1$. **[10]**

Solution

(a) Using

$$\cosh x = \frac{e^x + e^{-x}}{2} \qquad \text{and} \qquad \sinh x = \frac{e^x - e^{-x}}{2}$$

we have

$$\begin{aligned} 3\cosh x - \sinh x &= 3\left(\frac{e^x + e^{-x}}{2}\right) - \left(\frac{e^x - e^{-x}}{2}\right) \\ &= \frac{3e^x + 3e^{-x} - e^x + e^{-x}}{2} \\ &= \frac{2e^x + 4e^{-x}}{2} \\ &= e^x + 2e^{-x} \end{aligned}$$

(b) From part (a):

$$6(3\cosh x - \sinh x) = 6(e^x + 2e^{-x}) = 6e^x + 12e^{-x}$$

so the differential equation becomes

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} - 2y = 6e^x + 12e^{-x}$$

First solve the homogeneous equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} - 2y = 0$$

The auxiliary equation is

$$\lambda^2 - \lambda - 2 = 0$$

Factorising:

$$(\lambda - 2)(\lambda + 1) = 0$$

So the roots are $\lambda = 2$ and $\lambda = -1$. Therefore the complementary function is

$$y_c = Ae^{2x} + Be^{-x}$$

Now find a particular integral. Since the right-hand side is $6e^x + 12e^{-x}$, take

$$y_p = ae^x + bxe^{-x}$$

We use bxe^{-x} rather than be^{-x} because e^{-x} is already part of the complementary function.

For the ae^x term:

$$\frac{\mathrm{d}}{\mathrm{d}x}(ae^x) = ae^x, \qquad \frac{\mathrm{d}^2}{\mathrm{d}x^2}(ae^x) = ae^x$$

Substituting into the left-hand side:

$$\begin{aligned}\frac{\mathrm{d}^2}{\mathrm{d}x^2}(ae^x) - \frac{\mathrm{d}}{\mathrm{d}x}(ae^x) - 2(ae^x) &= ae^x - ae^x - 2ae^x \\ &= -2ae^x\end{aligned}$$

This must equal $6e^x$, so

$$-2a = 6$$

Hence

$$a = -3$$

For the bxe^{-x} term:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}(bxe^{-x}) &= b\frac{\mathrm{d}}{\mathrm{d}x}(xe^{-x}) \\ &= b\big(e^{-x} - xe^{-x}\big) \\ &= b(1-x)e^{-x}\end{aligned}$$

Differentiate again:

$$\begin{aligned}\frac{\mathrm{d}^2}{\mathrm{d}x^2}(bxe^{-x}) &= b\frac{\mathrm{d}}{\mathrm{d}x}\big((1-x)e^{-x}\big) \\ &= b\big((-1)e^{-x} + (1-x)(-e^{-x})\big) \\ &= b(x-2)e^{-x}\end{aligned}$$

Substituting into the left-hand side:

$$\begin{aligned}\frac{\mathrm{d}^2}{\mathrm{d}x^2}(bxe^{-x}) - \frac{\mathrm{d}}{\mathrm{d}x}(bxe^{-x}) - 2(bxe^{-x}) &= b(x-2)e^{-x} - b(1-x)e^{-x} - 2bxe^{-x} \\ &= b\big(x - 2 - 1 + x - 2x\big)e^{-x} \\ &= -3be^{-x}\end{aligned}$$

This must equal $12e^{-x}$, so

$$-3b = 12$$

Hence

$$b = -4$$

Therefore a particular integral is

$$y_p = -3e^x - 4xe^{-x}$$

and the general solution is

$$y = Ae^{2x} + Be^{-x} - 3e^x - 4xe^{-x}$$

Now use the initial conditions.

When $x = 0$, $y = 1$:

$$\begin{aligned} 1 &= Ae^0 + Be^0 - 3e^0 - 4(0)e^0 \\ &= A + B - 3 \end{aligned}$$

So

$$A + B = 4$$

Differentiate the general solution:

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}x} &= 2Ae^{2x} - Be^{-x} - 3e^x - 4\frac{\mathrm{d}}{\mathrm{d}x}(xe^{-x}) \\ &= 2Ae^{2x} - Be^{-x} - 3e^x - 4(1-x)e^{-x} \end{aligned}$$

When $x = 0$, $\frac{\mathrm{d}y}{\mathrm{d}x} = 1$:

$$\begin{aligned} 1 &= 2Ae^0 - Be^0 - 3e^0 - 4(1-0)e^0 \\ &= 2A - B - 3 - 4 \\ &= 2A - B - 7 \end{aligned}$$

So

$$2A - B = 8$$

Solve the simultaneous equations

$$\begin{aligned} A + B &= 4 \\ 2A - B &= 8 \end{aligned}$$

Adding gives

$$3A = 12$$

so

$$A = 4$$

Then

$$B = 4 - A = 0$$

Substituting back, the required solution is

$$y = 4e^{2x} - 3e^x - 4xe^{-x}$$

12. Find the particular solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 4\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = xe^{2x}$$

given that, when $x = 0$, $y = 1$ and $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$. **[10]**

Solution

The homogeneous equation is

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 4\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 0$$

The auxiliary equation is

$$\lambda^2 - 4\lambda + 4 = 0$$

So

$$(\lambda - 2)^2 = 0$$

Hence the complementary function is

$$y_c = (A + Bx)e^{2x}$$

For a particular integral, since the right-hand side is xe^{2x} and $\lambda = 2$ is a repeated root, try

$$y_p = e^{2x}(ax^3 + bx^2)$$

Then

$$\begin{aligned}\frac{\mathrm{d}y_p}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[e^{2x}(ax^3 + bx^2)\right] \\ &= 2e^{2x}(ax^3 + bx^2) + e^{2x}(3ax^2 + 2bx) \\ &= e^{2x}\left(2ax^3 + (2b + 3a)x^2 + 2bx\right)\end{aligned}$$

Differentiate again:

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[e^{2x}\left(2ax^3 + (2b + 3a)x^2 + 2bx\right)\right] \\ &= 2e^{2x}\left(2ax^3 + (2b + 3a)x^2 + 2bx\right) \\ &\quad + e^{2x}\left(6ax^2 + 2(2b + 3a)x + 2b\right) \\ &= e^{2x}\left(4ax^3 + (4b + 12a)x^2 + (8b + 6a)x + 2b\right)\end{aligned}$$

Substituting into the differential equation gives

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} - 4\frac{\mathrm{d}y_p}{\mathrm{d}x} + 4y_p &= e^{2x}\left(4ax^3 + (4b + 12a)x^2 + (8b + 6a)x + 2b\right) \\ &\quad - 4e^{2x}\left(2ax^3 + (2b + 3a)x^2 + 2bx\right) \\ &\quad + 4e^{2x}(ax^3 + bx^2)\end{aligned}$$

So

$$\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} - 4\frac{\mathrm{d}y_p}{\mathrm{d}x} + 4y_p = e^{2x}(6ax + 2b)$$

Equating coefficients with xe^{2x} gives

$$6a = 1, \qquad 2b = 0$$

So

$$a = \frac{1}{6}, \qquad b = 0$$

Therefore

$$y_p = \frac{x^3}{6}e^{2x}$$

Thus the general solution is

$$y = (A + Bx)e^{2x} + \frac{x^3}{6}e^{2x}$$

So

$$y = e^{2x}\left(A + Bx + \frac{x^3}{6}\right)$$

Now use the initial conditions.

When $x = 0$, $y = 1$:

$$\begin{aligned} 1 &= e^0\left(A + B \times 0 + \frac{0^3}{6}\right) \\ &= A \end{aligned}$$

So

$$A = 1$$

Now differentiate:

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[e^{2x}\left(A + Bx + \frac{x^3}{6}\right)\right] \\ &= 2e^{2x}\left(A + Bx + \frac{x^3}{6}\right) + e^{2x}\left(B + \frac{x^2}{2}\right) \\ &= e^{2x}\left(2A + 2Bx + \frac{x^3}{3} + B + \frac{x^2}{2}\right) \end{aligned}$$

When $x = 0$, $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$, so

$$0 = 2A + B$$

With $A = 1$:

$$0 = 2 + B$$

Hence

$$B = -2$$

Substituting $A = 1$ and $B = -2$, the required particular solution is

$$y = e^{2x}\left(\frac{x^3}{6} - 2x + 1\right)$$